

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Dataset Integration

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO3: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

Assessment Tool Weightage: 30%

Total Marks: 20

Code of Conduct

I **Yarra Pawan (192424294)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Y. Pawan**

Dataset Integration Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Dataset Understanding & Preparation	30%	Correctly understands both datasets, identifies relevant columns/keys, and loads them properly.	Minor misunderstanding or preparation issue.	Loads datasets but needs some guidance in identifying relevant fields.	Unable to identify or prepare the datasets correctly.
Dataset Integration	20%	Correctly integrates datasets using appropriate merge(), join(), or concat() and selects the correct integration key.	Integration is mostly correct with minor errors.	Performs integration with significant guidance or minor key/operation errors.	Unable to integrate the datasets correctly.
Data Processing & Analysis	15%	Correctly performs all required calculations, filtering, grouping, or transformations on the integrated data.	Performs most required analysis correctly.	Performs basic analysis but misses some required operations.	Analysis is incorrect or incomplete.

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Python/Pandas Implementation	20%	Code is correct, efficient, well-structured, and uses appropriate Pandas functions.	Code works with minor inefficiencies/errors.	Code works partially and requires corrections.	Code contains major errors or does not execute.
Output & Interpretation	15%	Clearly presents the required output and correctly interprets the results.	Output is correct with limited interpretation.	Output is partially correct or poorly presented.	Output is missing/incorrect and results are not interpreted.

Total Marks Awarded:

Signature of the Course Faculty

Questions:

Set 1 – Dataset Integration Tasks

1. Employee–Department Integration

You are given two datasets:

- employees.csv — Employee_ID, Employee_Name, Department_ID, Salary
- departments.csv — Department_ID, Department_Name, Location

Task: Write a Python program using Pandas to integrate the two datasets based on Department_ID and create a new dataset containing employee name, department name, location, and salary.

2. Customer–Transaction Integration

You are given:

- customers.csv — Customer_ID, Customer_Name, City
- transactions.csv — Transaction_ID, Customer_ID, Product, Amount

Task: Integrate the datasets using Customer_ID. Display the customer name, city, product, and transaction amount. Identify customers who have made transactions above ₹10,000.

3. Student–Marks Integration

You are given:

- students.csv — Student_ID, Student_Name, Department

- marks.csv — Student_ID, Python, DBMS, Statistics

Task: Merge the datasets using Student_ID. Create a new column Total_Marks and identify students whose total marks are greater than 240.

4. Product–Sales Integration

You are given:

- products.csv — Product_ID, Product_Name, Category, Unit_Price
- sales.csv — Sale_ID, Product_ID, Quantity, Sale_Date

Task: Integrate the two datasets using Product_ID. Calculate the total sales amount for each transaction using:

$$\text{Total_Sales} = \text{Quantity} \times \text{Unit_Price}$$

Display the integrated dataset.

5. Hospital Patient–Visit Integration

You are given:

- patients.csv — Patient_ID, Patient_Name, Age, Gender
- visits.csv — Visit_ID, Patient_ID, Doctor, Visit_Date, Fee

Task: Integrate the patient and visit datasets using Patient_ID. Display patient name, age, doctor, visit date, and fee. Calculate the total consultation fee paid by each patient.

Set 2 – Dataset Integration Tasks

1. Customer–Order Integration

You are given:

- customer.csv — Customer_ID, Customer_Name, City
- orders.csv — Order_ID, Customer_ID, Order_Date, Order_Amount

Task: Merge the datasets using Customer_ID. Display customer name, city, order date, and order amount. Find the customer with the highest total order amount.

2. Employee–Attendance Integration

You are given:

- employee.csv — Employee_ID, Employee_Name, Department

- attendance.csv — Employee_ID, Date, Status

Task: Integrate the two datasets using Employee_ID. Calculate the number of days each employee was present and absent.

3. Course–Student Integration

You are given:

- students.csv — Student_ID, Student_Name, Department
- enrollment.csv — Student_ID, Course_ID, Course_Name, Credits

Task: Merge the datasets using Student_ID. Display student name, department, course name, and credits. Find the total number of credits registered by each student.

4. Sales–Region Integration

You are given:

- sales.csv — Sale_ID, Product_ID, Region_ID, Quantity, Amount
- regions.csv — Region_ID, Region_Name, Manager

Task: Integrate the datasets using Region_ID. Display the region name, manager, product ID, quantity, and amount. Calculate the total sales amount for each region.

5. Weather–City Integration

You are given:

- cities.csv — City_ID, City_Name, State
- weather.csv — City_ID, Date, Temperature, Humidity, Rainfall

Task: Merge the datasets using City_ID. Display city name, state, date, temperature, humidity, and rainfall. Identify the city with the highest average temperature.

1. Employee-Department Integration

Datasets:

employees.csv:

```
Employee_ID,Employee_Name,Department_ID,Salary
E001,Ravi,D001,45000
E002,Anita,D002,50000
E003,Kumar,D001,48000
E004,Priya,D003,55000
E005,Arun,D002,47000
```

departments.csv:

```
Department_ID,Department_Name,Location
D001,Computer Science,Chennai
D002,Human Resources,Bangalore
D003,Finance,Hyderabad
```

Python Code:

```
import pandas as pd
employees = pd.read_csv("employees.csv")
departments = pd.read_csv("departments.csv")
merged = pd.merge(
    employees,
    departments,
    on="Department_ID"
)
result = merged[
    ["Employee_Name", "Department_Name", "Location", "Salary"]
]
print("Employee-Department Integration:")
print(result)
result.to_csv("employee_department.csv", index=False)
```

Output:

The screenshot displays a code editor with a Python script named `q1.py` and its output in a terminal window. The script uses `pandas` to read `employees.csv` and `departments.csv`, merge them on `Department_ID`, and save the result to `employee_department.csv`. The terminal output shows the merged data as a table.

```
1 import pandas as pd
2 employees = pd.read_csv("employees.csv")
3 departments = pd.read_csv("departments.csv")
4 merged = pd.merge(
5     employees,
6     departments,
7     on="Department_ID"
8 )
9 result = merged[
10     ["Employee_Name", "Department_Name", "Location", "Salary"]
11 ]
12 print("Employee-Department Integration:")
13 print(result)
14 result.to_csv("employee_department.csv", index=False)
```

	Employee_Name	Department_Name	Location	Salary
0	Ravi	Computer Science	Chennai	45000
1	Anita	Human Resources	Bangalore	50000
2	Kumar	Computer Science	Chennai	48000
3	Priya	Finance	Hyderabad	55000
4	Arun	Human Resources	Bangalore	47000

2. Customer -Transaction Integration

customers.csv:

Customer_ID, Customer_Name, City
C001, Ravi, Chennai
C002, Anita, Bangalore
C003, Kumar, Hyderabad
C004, Priya, Mumbai
C005, Arun, Delhi

transactions.csv:

Transaction_ID, Customer_ID, Product, Amount
T001, C001, Laptop, 55000
T002, C002, Mobile, 18000
T003, C001, Mouse, 1500
T004, C003, Television, 35000
T005, C004, Keyboard, 2500

T006,C002,Headphones,4500

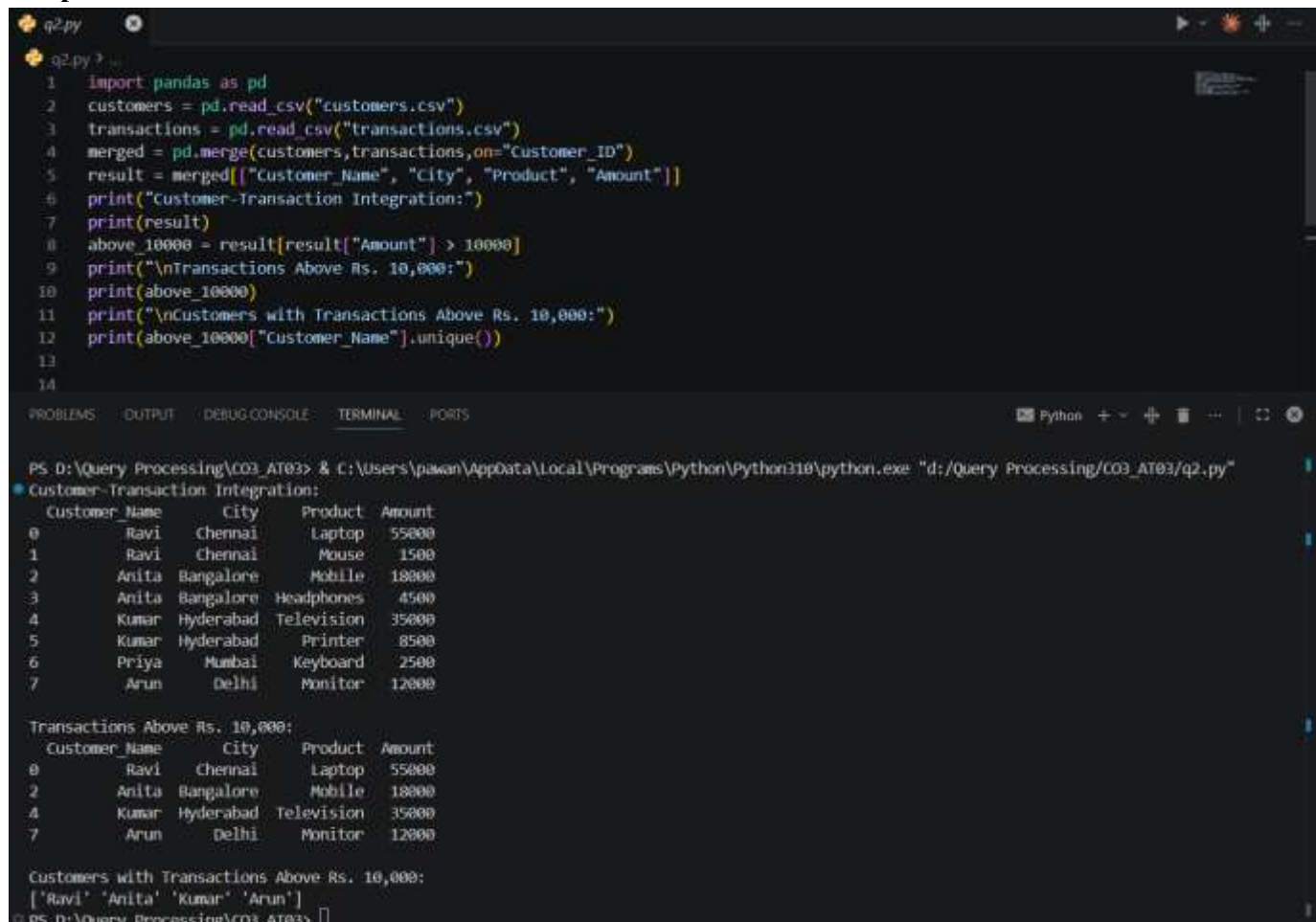
T007,C005,Monitor,12000

T008,C003,Printer,8500

Python Code:

```
import pandas as pd
customers = pd.read_csv("customers.csv")
transactions = pd.read_csv("transactions.csv")
merged = pd.merge(customers,transactions,on="Customer_ID")
result = merged[["Customer_Name", "City", "Product", "Amount"]]
print("Customer-Transaction Integration:")
print(result)
above_10000 = result[result["Amount"] > 10000]
print("\nTransactions Above Rs. 10,000:")
print(above_10000)
print("\nCustomers with Transactions Above Rs. 10,000:")
print(above_10000["Customer_Name"].unique())
```

Output:



```
q2.py
1 import pandas as pd
2 customers = pd.read_csv("customers.csv")
3 transactions = pd.read_csv("transactions.csv")
4 merged = pd.merge(customers,transactions,on="Customer_ID")
5 result = merged[["Customer_Name", "City", "Product", "Amount"]]
6 print("Customer-Transaction Integration:")
7 print(result)
8 above_10000 = result[result["Amount"] > 10000]
9 print("\nTransactions Above Rs. 10,000:")
10 print(above_10000)
11 print("\nCustomers with Transactions Above Rs. 10,000:")
12 print(above_10000["Customer_Name"].unique())
13
14
```

PS D:\Query Processing\CO3_AT03> & C:\Users\pawan\AppData\Local\Programs\Python\Python310\python.exe "d:\Query Processing\CO3_AT03\q2.py"

Customer-Transaction Integration:

	Customer_Name	City	Product	Amount
0	Ravi	Chennai	Laptop	55000
1	Ravi	Chennai	Mouse	1500
2	Anita	Bangalore	Mobile	18000
3	Anita	Bangalore	Headphones	4500
4	Kumar	Hyderabad	Television	35000
5	Kumar	Hyderabad	Printer	8500
6	Priya	Mumbai	Keyboard	2500
7	Arun	Delhi	Monitor	12000

Transactions Above Rs. 10,000:

	Customer_Name	City	Product	Amount
0	Ravi	Chennai	Laptop	55000
2	Anita	Bangalore	Mobile	18000
4	Kumar	Hyderabad	Television	35000
7	Arun	Delhi	Monitor	12000

Customers with Transactions Above Rs. 10,000:

```
['Ravi' 'Anita' 'Kumar' 'Arun']
```

PS D:\Query Processing\CO3_AT03>

3. Student-Marks Integration

students.csv:

```
Student_ID,Student_Name,Department
S001,Ravi,CSE
S002,Anita,ECE
S003,Kumar,IT
S004,Priya,CSE
S005,Arun,EEE
```

marks.csv:

```
Student_ID,Python,DBMS,Statistics
S001,85,90,88
S002,75,80,82
S003,95,92,90
S004,88,85,91
S005,70,75,78
```

Python Code:

```
import pandas as pd
students = pd.read_csv("students.csv")
marks = pd.read_csv("marks.csv")
merged = pd.merge(students,marks,on="Student_ID")
merged["Total_Marks"] = (merged["Python"] + merged["DBMS"] + merged["Statistics"])
print("Student-Marks Integration:")
print(merged)
high_scorers = merged[
    merged["Total_Marks"] > 240
]

print("\nStudents with Total Marks Greater Than 240:")
print(
    high_scorers[
        ["Student_Name", "Department", "Total_Marks"]
    ]
)
```


Output:

The screenshot shows a Visual Studio Code window with a file named `q3.py`. The code uses pandas to merge two CSV files, filter students based on total marks, and print the results.

```
1 import pandas as pd
2 students = pd.read_csv("students.csv")
3 marks = pd.read_csv("marks.csv")
4 merged = pd.merge(students, marks, on="Student_ID")
5 merged["Total_Marks"] = (merged["Python"] + merged["DBMS"] + merged["Statistics"])
6 print("Student-Marks Integration:")
7 print(merged)
8 high_scorers = merged[
9     merged["Total_Marks"] > 240
10 ]
11
12 print("\nStudents with Total Marks Greater Than 240:")
13 print(
14     high_scorers[
15         ["Student_Name", "Department", "Total_Marks"]
16     ]
17 )
```

The bottom panel shows the terminal output:

```
PS D:\Query Processing\CO3_AT03> & C:\Users\pawan\AppData\Local\Programs\Python\Python310\python.exe "d:/Query Processing/CO3_AT03/q3.py"
```

```
• Student-Marks Integration:
```

	Student_ID	Student_Name	Department	Python	DBMS	Statistics	Total_Marks
0	S001	Ravi	CSE	85	90	88	263
1	S002	Anita	ECE	75	80	82	237
2	S003	Kumar	IT	95	92	90	277
3	S004	Priya	CSE	88	85	91	264
4	S005	Arun	EEE	70	75	78	223


```
Students with Total Marks Greater Than 240:
```

	Student_Name	Department	Total_Marks
0	Ravi	CSE	263
2	Kumar	IT	277
3	Priya	CSE	264

```
• PS D:\Query Processing\CO3_AT03>
```

4. Product-Sales Integration

products.csv:

```
Product_ID,Product_Name,Category,Unit_Price
P001,Laptop,Electronics,55000
P002,Mobile,Electronics,18000
P003,Keyboard,Accessories,2500
P004,Mouse,Accessories,1200
P005,Monitor,Electronics,15000
```

sales.csv:

```
Sale_ID,Product_ID,Quantity,Sale_Date
SA001,P001,2,2026-08-01
SA002,P002,3,2026-08-02
SA003,P003,5,2026-08-03
SA004,P004,10,2026-08-04
SA005,P005,2,2026-08-05
```

Python Code:

```
import pandas as pd
products = pd.read_csv("products.csv")
sales = pd.read_csv("sales.csv")
merged = pd.merge(sales,products,on="Product_ID")
merged["Total_Sales"] = (merged["Quantity"] * merged["Unit_Price"])
result =
merged[["Sale_ID","Product_ID","Product_Name","Category","Unit_Price","Quantity","Sale_Date","Total_Sales"]]
print("Product-Sales Integration:")
print(result)
result.to_csv("product_sales_integrated.csv", index=False)
```

Output:

The screenshot displays a Python IDE environment with three main components: a CSV viewer, a Python script editor, and a terminal window.

CSV Viewer (product_sales_integrated.csv):

	A	B	C	D	E	F	G	H
	Sale_ID	Product_ID	Product_Name	Category	Unit_Price	Quantity	Sale_Date	Total_Sales
1	SA001	P001	Laptop	Electro...	55000	2	2026...	110000
2	SA002	P002	Mobile	Electro...	18000	3	2026...	54000
3	SA003	P003	Keyboard	Access...	2500	5	2026...	12500
4	SA004	P004	Mouse	Access...	1200	10	2026...	12000
5	SA005	P005	Monitor	Electro...	15000	2	2026...	30000

Python Script (q4.py):

```
1 import pandas as pd
2 products = pd.read_csv("products.csv")
3 sales = pd.read_csv("sales.csv")
4 merged = pd.merge(sales, products, on="Product_ID")
5 merged["Total_Sales"] = (merged["Quantity"] * merged["Unit_Price"])
6 result = merged[["Sale_ID", "Product_ID", "Product_Name", "Total_Sales"]]
7 print("Product-Sales Integration:")
8 print(result)
9 result.to_csv("product_sales_integrated.csv", index=False)
```

Terminal Output:

```
PS D:\Query Processing\CO3_AT03> & C:\Users\pawan\AppData\Local\Programs\Python\Python310\python.exe "d:/Query Processing/CO3_AT03/q4.py"
Product-Sales Integration:
  Sale_ID Product_ID Product_Name  Category  Unit_Price  Quantity  Sale_Date  Total_Sales
0  SA001      P001      Laptop  Electronics      55000         2  2026-08-01      110000
1  SA002      P002      Mobile  Electronics      18000         3  2026-08-02       54000
2  SA003      P003   Keyboard  Accessories       2500         5  2026-08-03      12500
3  SA004      P004      Mouse  Accessories       1200        10  2026-08-04      12000
4  SA005      P005     Monitor  Electronics      15000         2  2026-08-05      30000
PS D:\Query Processing\CO3_AT03>
```

5. Hospital Patient–Visit Integration

patients.csv:

```
Patient_ID,Patient_Name,Age,Gender
P001,Ravi,25,Male
P002,Anita,32,Female
P003,Kumar,45,Male
P004,Priya,28,Female
P005,Arun,51,Male
```

Create visits.csv:

```
Visit_ID,Patient_ID,Doctor,Visit_Date,Fee
V001,P001,Dr. Kumar,2026-08-01,500
V002,P002,Dr. Priya,2026-08-01,700
V003,P001,Dr. Kumar,2026-08-05,500
V004,P003,Dr. Ravi,2026-08-03,1000
V005,P004,Dr. Anita,2026-08-04,800
V006,P002,Dr. Priya,2026-08-06,700
V007,P005,Dr. Kumar,2026-08-07,1200
V008,P003,Dr. Ravi,2026-08-08,1000
```

Python Code:

```
import pandas as pd

patients = pd.read_csv("patients.csv")
visits = pd.read_csv("visits.csv")

merged = pd.merge(patients,visits,on="Patient_ID")
result = merged[["Patient_Name","Age","Doctor","Visit_Date","Fee"]]
print("Hospital Patient-Visit Integration:")
print(result)

total_fee = merged.groupby(["Patient_ID", "Patient_Name"])["Fee"].sum()
print("\nTotal Consultation Fee Paid by Each Patient:")
print(total_fee)
```

Output:

```
q5.py
q5.py > ...
1 import pandas as pd
2
3 patients = pd.read_csv("patients.csv")
4 visits = pd.read_csv("visits.csv")
5
6 merged = pd.merge(patients,visits,on="Patient_ID")
7 result = merged[["Patient_Name","Age","Doctor","Visit_Date","Fee"]]
8 print("Hospital Patient-Visit Integration:")
9 print(result)
10
11 total_fee = merged.groupby(["Patient_ID", "Patient_Name"])[["Fee"]].sum()
12 print("\nTotal Consultation Fee Paid by Each Patient:")
13 print(total_fee)
```

PS D:\Query Processing\CO3_AT03> & C:\Users\pawan\AppData\Local\Programs\Python\Python310\python.exe "d:/Query Processing/CO3_AT03/q5.py"

Hospital Patient-Visit Integration:

	Patient_Name	Age	Doctor	Visit_Date	Fee
9	Ravi	25	Dr. Kumar	2026-08-01	500
1	Ravi	25	Dr. Kumar	2026-08-05	500
2	Anita	32	Dr. Priya	2026-08-01	700
3	Anita	32	Dr. Priya	2026-08-06	700
4	Kumar	45	Dr. Ravi	2026-08-03	1000
5	Kumar	45	Dr. Ravi	2026-08-08	1000
6	Priya	28	Dr. Anita	2026-08-04	800
7	Arun	51	Dr. Kumar	2026-08-07	1200

Total Consultation Fee Paid by Each Patient:

Patient_ID	Patient_Name	
P001	Ravi	1000
P002	Anita	1400
P003	Kumar	2000
P004	Priya	800
P005	Arun	1200

Name: Fee, dtype: int64

PS D:\Query Processing\CO3_AT03>